

TITLE: CREATING TWO-DIMENSIONAL OBJECTS (HOUSE, CAR, FISH, MAN) USING C GRAPHICS FUNCTIONS

OBJECTIVE:

- To create simple two-dimensional objects such as a house, car, fish, and man using basic graphics functions in C and to generate their pixel activation lists.

THEORY:

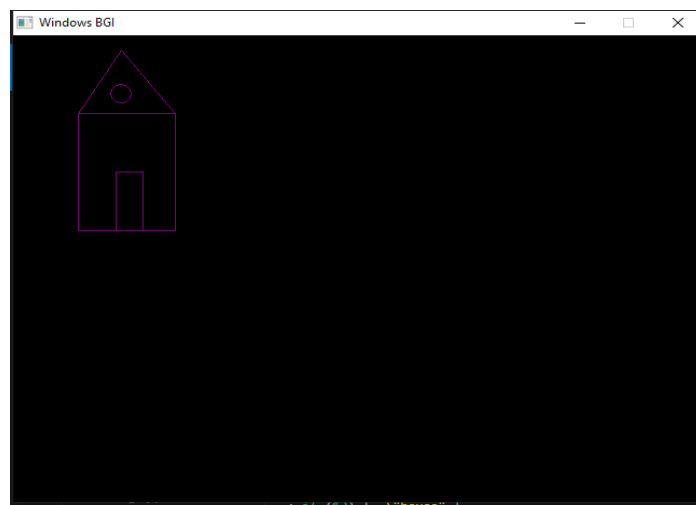
In computer graphics, 2D objects are drawn on the screen using pixel-based algorithms. C language provides several graphics functions for constructing basic shapes:

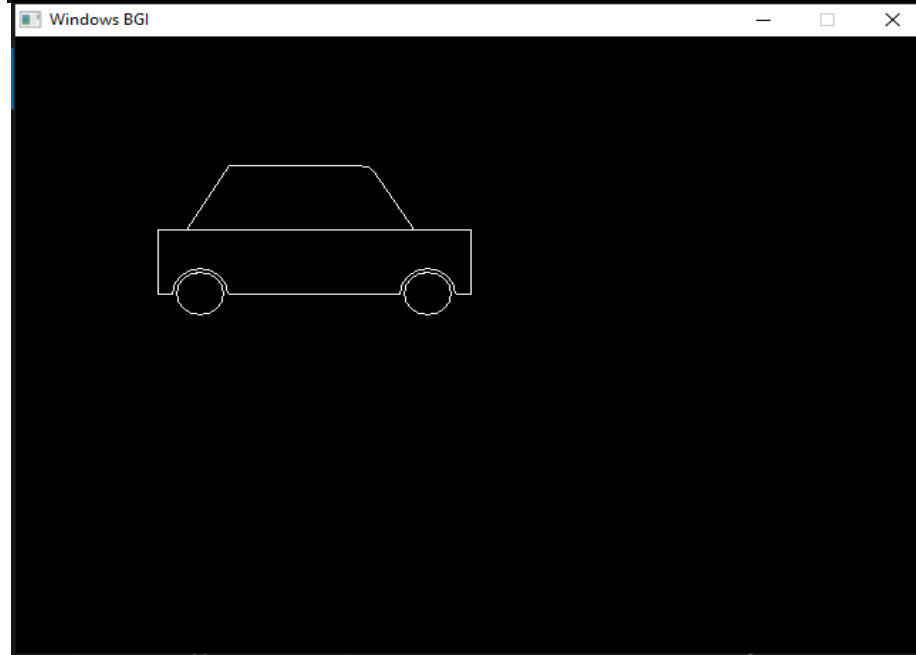
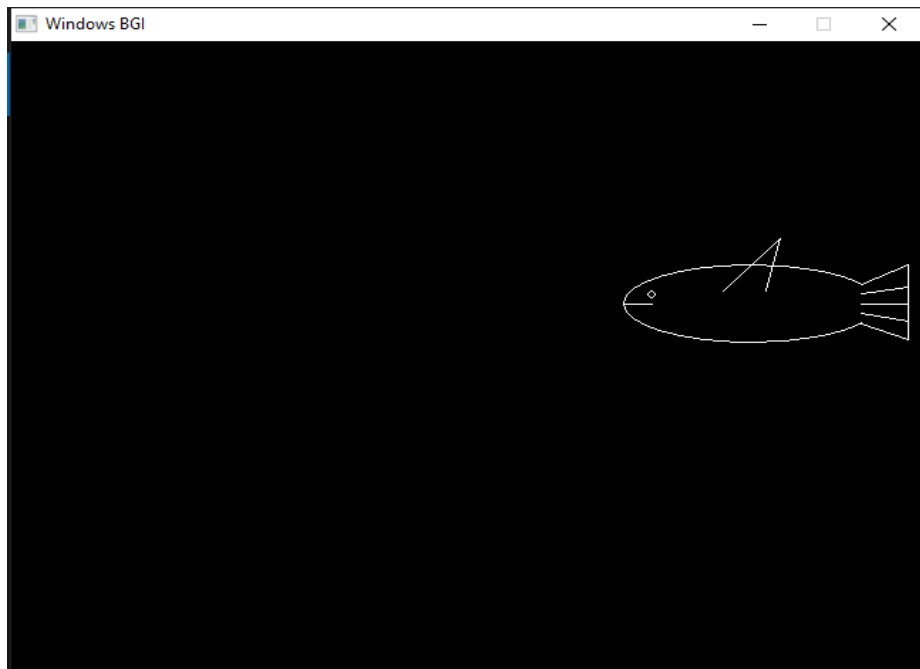
- `line(x1, y1, x2, y2)` – Draws a line from point (x1, y1) to (x2, y2).
- `circle(x, y, r)` – Draws a circle with center (x, y) and radius r.
- `ellipse(x, y, startangle, endangle, xradius, yradius)` – Draws an ellipse. For a complete ellipse, use `startangle = 0` and `endangle = 360`.
- `rectangle(left, top, right, bottom)` – Draws a rectangle using the top-left and bottom-right corners.
- `drawpoly(num, points[])` – Draws polygons using an array of coordinates. `num` = number of vertices + 1 (first point repeated at the end).

By combining these functions, we can construct complex 2D objects. For example:

1. House: Rectangle (base), triangle (roof), small rectangles (door and windows)
2. Car: Rectangle (body), polygon (roof), circles (wheels)
3. Fish: Ellipse (body), triangle (tail), circle (eye)
4. Man: Circle (head), lines (body, arms, legs)

OUTPUT:







CONCLUSION:

- The experiment demonstrated how basic C graphics functions can be combined to create complex 2D objects.
- Using lines, rectangles, circles, ellipses, and polygons, we can represent real-life objects in a 2D plane.
- The pixel activation list generated during execution helps visualize how graphics are drawn on a screen.
- The experiment provided practical understanding of coordinate systems, graphics functions, and object construction in C programming.